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☞ Chapter 3 Fundamentals of Fluid Motion

Chapter 3 Fundamentals of Fluid Motion (6 credit hours)

Content:

1. Eulerian and Lagrangian methods for describing fluid motion.
2. Steady, unsteady flow, uniform, and non-uniform flows; streamlines, pathlines, dye lines; 1D, 2D, and 3D flows.
3. Continuity equation.
4. Motion of fluid particles; rotational and irrotational fluid motion
5. Balance equations for ideal fluids (derived) and real fluids (not derived); Bernoulli integral for ideal fluids and energetic and geometrical meanings of Bernoulli's equation.
6. Potential function, stream function, flow net, and potential flow superposition.

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Chapter 3 Fundamentals of Fluid Motion

Chapter 3 Fundamentals of Fluid Motion

Section 1 How to describe fluid motion

1, Two ways to describe fluid motion

2, Euler acceleration

1, Two ways to describe fluid motion

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1. Lagrangian method

The Lagrangian method describes the motion of fluid particles through space and time. Their coordinates, motion over time, and interactions are used to describe the flow.

Spatial coordinates
$$\begin{cases} x = x(a,b,c,t) \\ y = y(a,b,c,t) \\ z = z(a,b,c,t) \end{cases}$$

(*a*, *b*, *c*) are the spatial coordinates of a fluid particle at starting time $t=t_0$, which is called the Lagrangian number. Therefore, the position **(*x*, *y*, *z*)** of any fluid particle in space can be regarded as a function of **(*a*, *b*, *c*)** and time ***t***.

(not used in this course)

2. Eulerian method

The Eulerian method describes locations in space through which the fluid flows at given times rather than the motion of the fluid itself. Flow parameters, such as flow velocity, density, and pressure, at each spatial location at each time are used to describe the flow.

Flow velocity
$$\begin{cases} u_x = u_x(x, y, z, t) \\ u_y = u_y(x, y, z, t) \\ u_z = u_z(x, y, z, t) \end{cases}$$

(x,y,z,t)——Eulerian coordinates

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Chapter 3 Fundamentals of Fluid Motion

Section 1 How to describe fluid motion

1, Two ways to describe fluid motion

☞ 2, Euler acceleration

2, Euler acceleration

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The acceleration (time derivative of velocity) of a fluid particle has two components:

(1) Local acceleration——due to a change of local flow velocity over time

(2) Convective acceleration——due to a change of flow velocity in space

Since the position of a fluid particle is again a function of time ***t***, the acceleration can be obtained as:

$$a_x = \frac{\mathrm{d}u_x}{\mathrm{d}t} = \frac{\partial u_x}{\partial t} + \frac{\partial u_x}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial u_x}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial u_x}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}t}$$

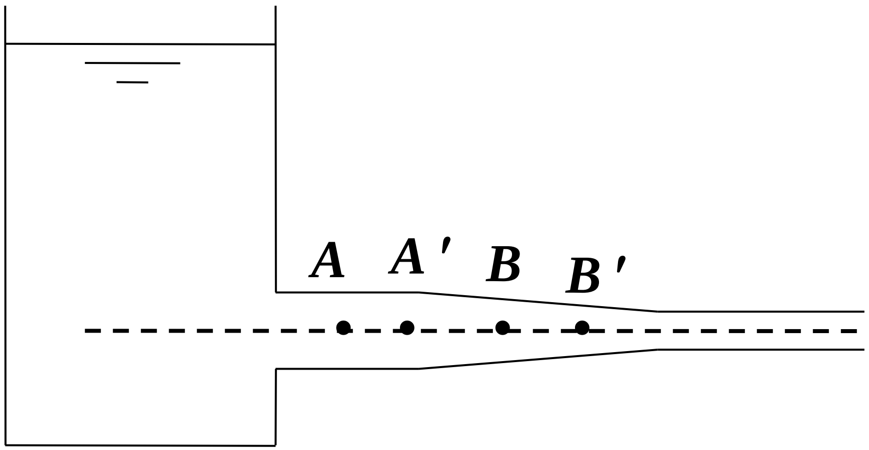
$$\because \frac{\mathrm{d}x}{\mathrm{d}t} = u_x, \frac{\mathrm{d}y}{\mathrm{d}t} = u_y, \frac{\mathrm{d}z}{\mathrm{d}t} = u_z$$

$$\therefore \begin{cases} a_x = \dfrac{\mathrm{d}u_x}{\mathrm{d}t} = \boxed{\dfrac{\partial u_x}{\partial t}} + \boxed{u_x \dfrac{\partial u_x}{\partial x} + u_y \dfrac{\partial u_x}{\partial y} + u_z \dfrac{\partial u_x}{\partial z}} \\ a_y = \dfrac{\mathrm{d}u_y}{\mathrm{d}t} = \boxed{\dfrac{\partial u_y}{\partial t}} + \boxed{u_x \dfrac{\partial u_y}{\partial x} + u_y \dfrac{\partial u_y}{\partial y} + u_z \dfrac{\partial u_y}{\partial z}} \\ a_z = \dfrac{\mathrm{d}u_z}{\mathrm{d}t} = \boxed{\dfrac{\partial u_z}{\partial t}} + \boxed{u_x \dfrac{\partial u_z}{\partial x} + u_y \dfrac{\partial u_z}{\partial y} + u_z \dfrac{\partial u_z}{\partial z}} \end{cases}$$

Local acceleration

Convective acceleration

Example: There are two points ***A*** and ***B*** in the water outlet pipe of a container. Analyze the acceleration of particles flowing through ***A*** and ***B*** when the water level of the container remains unchanged (constant) and the water level changes with time (not constant).



Assume that fluid particles at ***A*** and ***B*** moved to ***A′*** and ***B′***:

1, Constant water level:

(1) **A→A′** No local and convective acceleration

(2) **B→B′** No local, but convective acceleration

2, Changing water level:

(1) **A→A′** Local, but no convective acceleration

(2) **B→B′** Both local and convective acceleration

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Chapter 3 Fundamentals of Fluid Motion

Section 2 Basic concepts of fluid flow

1, Visualizations of flow

To visualize flow fields, the following methods are used:

1, Streamlines

2, Pathlines

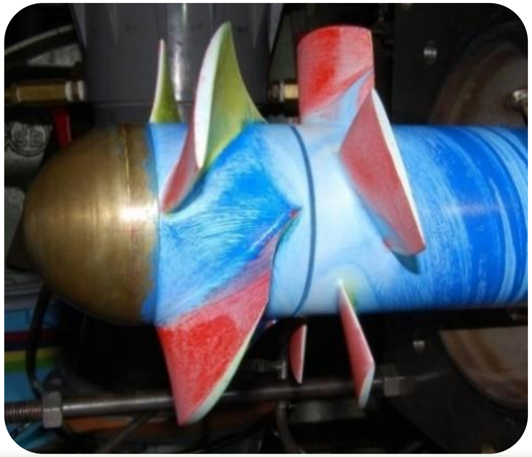
3, Dye lines

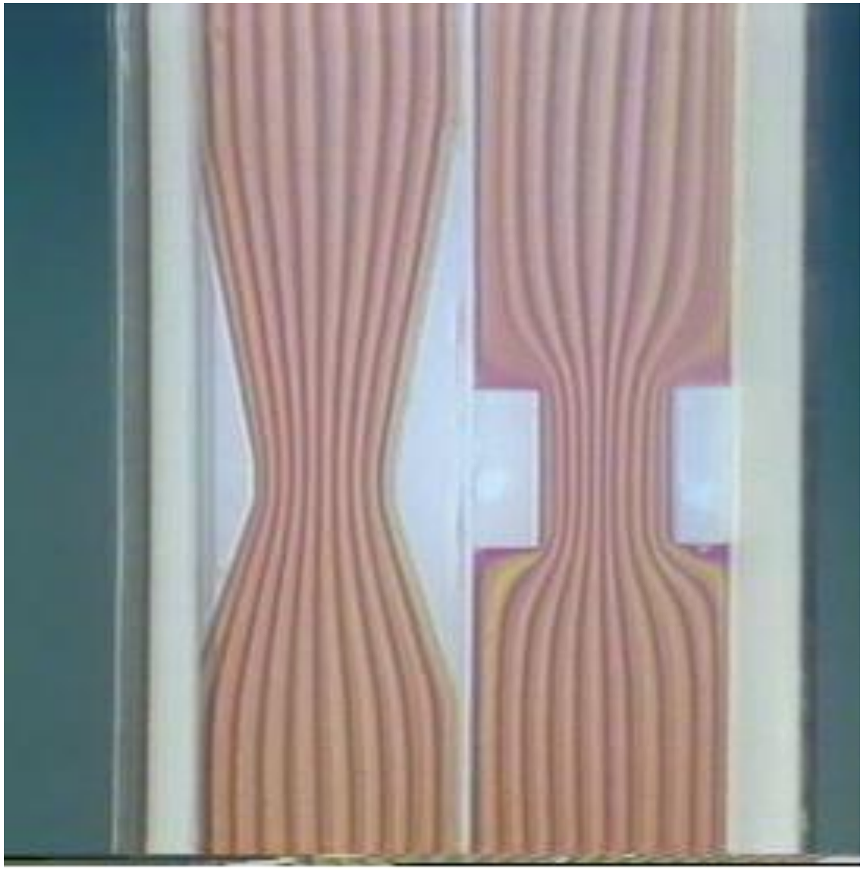
These three "lines" are not easy to observe in the actual flow field, but their shapes can be illustrated by experimental means and flow field display technologies such as tracer media.

1, Streamlines

(1) Definition

A stream line is a curve representing the flow trend of each point at a certain time. The tangent coincides with the flow direction. It describes the motion of different particles in the flow field at the same time.





流谱流线

Flow Spectrum And Streamline

1, Streamlines

(2) Drawing method

Pick any point in the flow field, draw the flow velocity vector of the fluid particle passing through this and a close neighboring point at a certain time.

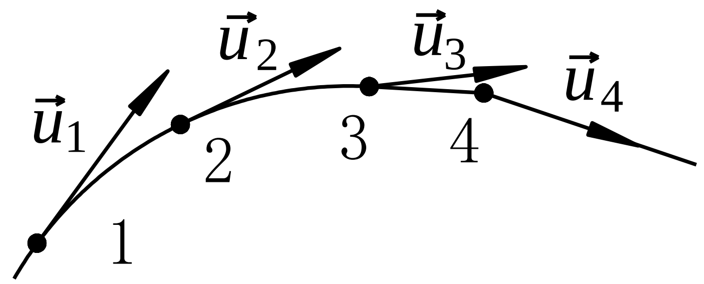


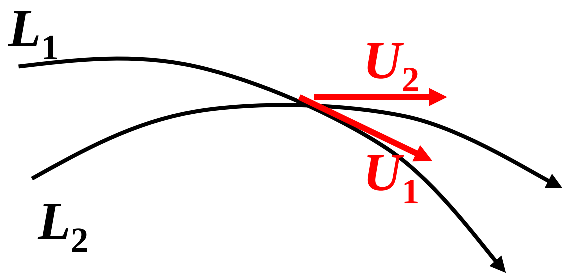
图3-6 某时刻流线图

If this continues, a broken line is obtained. If the points are infinitely close, the limit is the streamline at a certain time.

1, Streamlines

(3) Properties

a, Streamlines do not intersect



b, Streamlines are smooth curves

流谱流线

Flow Spectrum And Streamline

c, For incompressible fluids, the density of streamline clusters reflects the size of the flow velocity (the velocity is high where the streamlines are dense and low where they are sparse).

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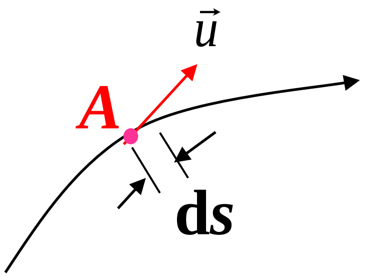
1, Streamlines

(4) Streamline equations



Line element: $\mathrm{d}\vec{s} = \mathrm{d}x\vec{i} + \mathrm{d}y\vec{j} + \mathrm{d}z\vec{k}$

Velocity field: $\vec{u} = u_x\vec{i} + u_y\vec{j} + u_z\vec{k}$

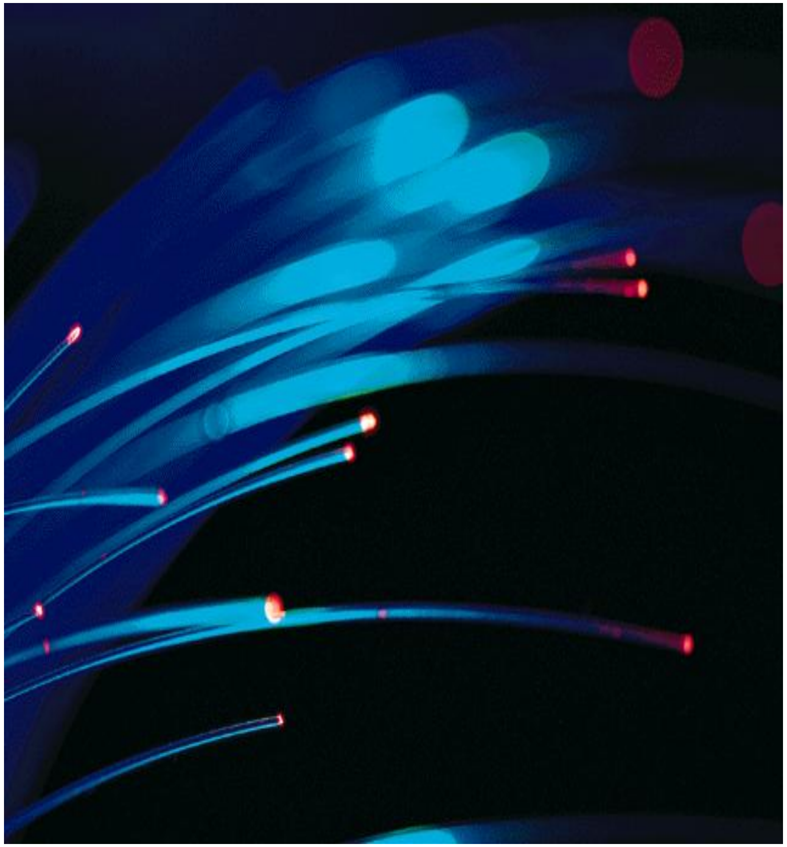
$\mathrm{d}\vec{s} \times \vec{u} = 0$

$$\frac{\mathrm{d}x}{u_\mathrm{x}} = \frac{\mathrm{d}y}{u_\mathrm{y}} = \frac{\mathrm{d}z}{u_\mathrm{z}}$$

2. Pathlines

(1), Definition

Trajectories of fluid particles as functions of time.



(2), Pathline equations

$$\frac{dx}{u_x} = \frac{dy}{u_y} = \frac{dz}{u_z} = dt$$

Here, $\boldsymbol{u_x}$, $\boldsymbol{u_y}$, $\boldsymbol{u_z}$ are all functions of space and time $\boldsymbol{t}$, which is an independent variable.

Differences between streamline and pathline equations:

$$\frac{dx}{u_x} = \frac{dy}{u_y} = \frac{dz}{u_z}$$

—— in streamline equations, $\boldsymbol{t}$ is given

3, Dye lines (also known as streaklines)

Connection of many fluid particle trajectories passing through a given point in the past.

Streamlines, pathlines, dye lines

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Lines	Definition	Notes
Streamlines	Streamlines are curves representing the flow velocity (or fluid particle velocity) direction at each point at a given time.	Equation: $$\frac{\mathrm{d}x}{u_{\mathrm{x}}}=\frac{\mathrm{d}y}{u_{\mathrm{y}}}=\frac{\mathrm{d}z}{u_{\mathrm{z}}}$$ Time ***t*** is given
Pathlines	Pathlines are the trajectories of fluid particles as functions of time.	Equation: $$\frac{\mathrm{d}x}{u_{\mathrm{x}}}=\frac{\mathrm{d}y}{u_{\mathrm{y}}}=\frac{\mathrm{d}z}{u_{\mathrm{z}}}=\mathrm{d}t$$ ***t*** is an independent variable
Dye lines	Connection of all fluid particle trajectories passing through a given point in the past.	In steady flow, streamlines, pathlines, and dye lines coincide.

Example 1 Velocity field: $u_x = \frac{x}{1+t}$, $u_y = y$, $u_z = 0$

Find streamline and pathline going through (***x*~0~**, ***y*~0~**, ***z*~0~**) at time ***t*~0~**.

(1) Streamline
$$\frac{\mathrm{d}x}{u_\mathrm{x}} = \frac{\mathrm{d}y}{u_\mathrm{y}} = \frac{\mathrm{d}z}{u_\mathrm{z}}$$

$$\frac{1+t}{x}\mathrm{d}x = \frac{1}{y}\mathrm{d}y$$

$(1+t)\ln x = \ln y + \ln C$

$$y = Cx^{(1+t)}$$

$t = t_0$, $x = x_0$, $y = y_0$ $\quad C = y_0 x_0^{-(1+t_0)}$

$$y = y_0 x_0^{-(1+t_0)} x^{(1+t)}$$

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Example 1 Velocity field: $u_x = \dfrac{x}{1+t},\quad u_y = y,\quad u_z = 0$

Find streamline and pathline going through (***x***$_0$, ***y***$_0$, ***z***$_0$) at time ***t***$_0$.

(2) Pathlines: $\dfrac{\mathrm{d}x}{u_\mathrm{x}} = \dfrac{\mathrm{d}y}{u_\mathrm{y}} = \mathrm{d}t$

$$\begin{cases} \dfrac{\mathrm{d}x}{x/1+t} = \mathrm{d}t \\ \dfrac{\mathrm{d}y}{y} = \mathrm{d}t \end{cases} \Rightarrow \begin{cases} \dfrac{\mathrm{d}x}{x} = \dfrac{\mathrm{d}t}{1+t} \\ \dfrac{\mathrm{d}y}{y} = \mathrm{d}t \end{cases} \Rightarrow \begin{cases} \ln x = \ln(1+t) + \ln C_1 \\ \ln y = \ln e^t + \ln C_2 \end{cases}$$

$$\begin{cases} x = C_1(1+t) \\ y = C_2 e^t \end{cases} \xrightarrow[x = x_0,\, y = y_0]{t = t_0} \begin{cases} C_1 = \dfrac{x_0}{1+t_0} \\ C_2 = y_0 e^{-t_0} \end{cases} \Rightarrow \begin{cases} x = \dfrac{x_0}{1+t_0}(1+t) \\ y = y_0 e^{(t-t_0)} \end{cases}$$

t is an independent variable

$$y = \frac{y_0}{e^{t_0}}\, e^{\left(\frac{1+t_0}{x_0}x-1\right)}$$

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Example 2 Velocity field:

$$u_x = \frac{-Cy}{x^2+y^2}, u_y = \frac{Cx}{x^2+y^2}, u_z = 0$$

Find streamlines and pathlines (constant **C>0**).

Streamlines: $$\frac{dx}{u_x} = \frac{dy}{u_y} \qquad \frac{dx}{\frac{-Cy}{x^2+y^2}} = \frac{dy}{\frac{Cx}{x^2+y^2}}$$

$$\therefore \qquad -\frac{dx}{y} = \frac{dy}{x} \qquad x^2+y^2 = C_1$$

Pathlines: $$\frac{dx}{u_x} = \frac{dy}{u_y} = dt \qquad \frac{dx}{\frac{-Cy}{x^2+y^2}} = \frac{dy}{\frac{Cx}{x^2+y^2}}$$

$$x^2+y^2 = C_2$$

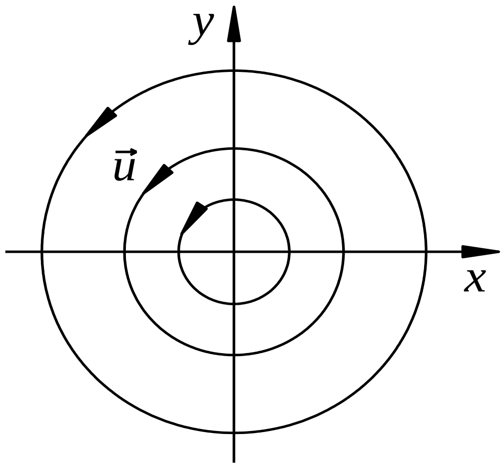


图3−11 速度场

Streamlines and pathlines are concentric circles

Direction is counterclockwise because for $y=0, x>0$, $u_x = 0, u_y > 0$

Example 3 Velocity field: $u_x = x + t, u_y = -y + t, u_z = 0$

Find streamline and pathline going through (**-1, -1, *z***) at time ***t*=0.**

(1) Streamline: $\dfrac{\mathrm{d}x}{u_x} = \dfrac{\mathrm{d}y}{u_y}$

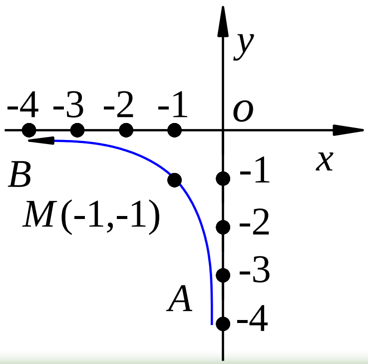
$$\frac{\mathrm{d}x}{x+t} = \frac{\mathrm{d}y}{-y+t}$$

$$\therefore \ln(x+t) = -\ln(-y+t) + \ln C$$

$$\therefore (x+t)(-y+t) = C$$

***t*=0, *x*=-1, *y*=-1 ->**

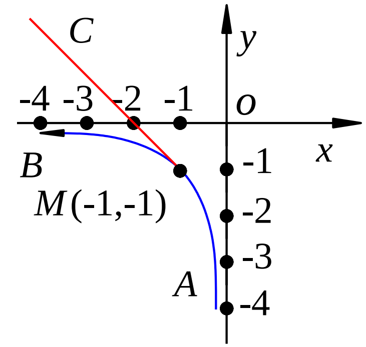
Streamline: *xy*=1



(2) Pathline: $\dfrac{\mathrm{d}x}{u_x} = \dfrac{\mathrm{d}y}{u_y} = \mathrm{d}t$

$$\frac{\mathrm{d}x}{x+t} = \frac{\mathrm{d}y}{-y+t} = \mathrm{d}t$$

$$\begin{cases} x = C_1 \mathrm{e}^{t} - t - 1 \\ y = C_2 \mathrm{e}^{-t} + t - 1 \end{cases}$$



***t*=0, *x*= –1, *y*=-1 -> *C*$_1$=0, *C*$_2$=0 ->**

Pathline: $\begin{cases} x = -t - 1 \\ y = t - 1 \end{cases}$

$$x + y + 2 = 0$$

Chapter 3 Fundamentals of Fluid Motion

Section 2 Basic concepts of fluid flow

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2, Euler method for classification of flows

1, Steady and unsteady flows

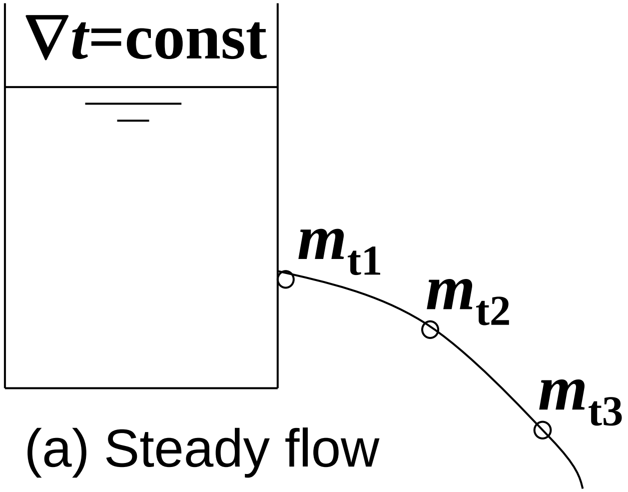
2, Uniform and non-uniform flows

3, Gradually-varied and rapidly-varied flows

4, 1D, 2D, and 3D flow

1, Steady and unsteady flow

Steady flow: flow parameters (***u***, ***p***, ***h***, etc.) at any spatial point do not change with time.



(a) Steady flow

$$\frac{\partial \vec{u}}{\partial t} = 0, \vec{u} = \vec{u}(x, y, z)$$

$$\frac{\partial p}{\partial t} = 0, p = p(x, y, z)$$

$$\frac{\partial u_x}{\partial t} = \frac{\partial u_y}{\partial t} = \frac{\partial u_z}{\partial t} = 0$$

Unsteady flow

(1) Definition

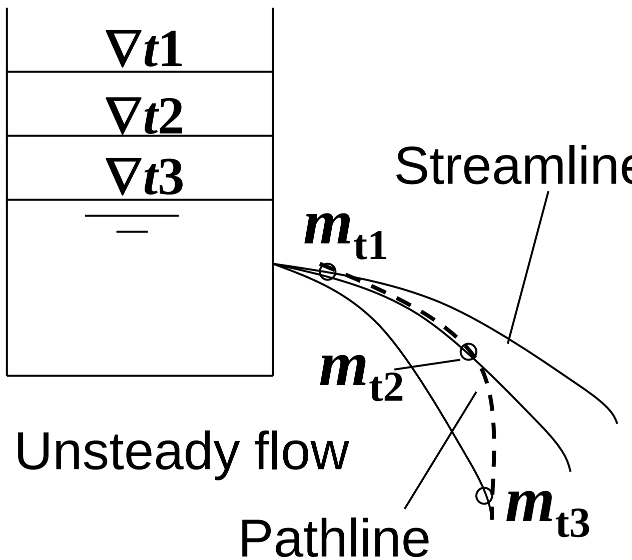
Unsteady flow: flow in which at least one flow parameter changes with time.

$$\frac{\partial \vec{u}}{\partial t} \neq 0, \vec{u} = \vec{u}(x, y, z, t)$$

$$\frac{\partial p}{\partial t} \neq 0, p = p(x, y, z, t)$$

$$\frac{\partial u_x}{\partial t}, \frac{\partial u_y}{\partial t}, \frac{\partial u_z}{\partial t}$$

At least one of the three is not equal to **0**.



(b) Unsteady flow

(2) Notes

For unsteady flow, the positions of streamlines vary with time; the streamlines do not coincide with the pathlines.

For steady flow, the positions of the streamlines do not change with time and coincide with the pathlines.

2, Euler method for classification of flows

1, Steady and unsteady flows

2, Uniform and non-uniform flows

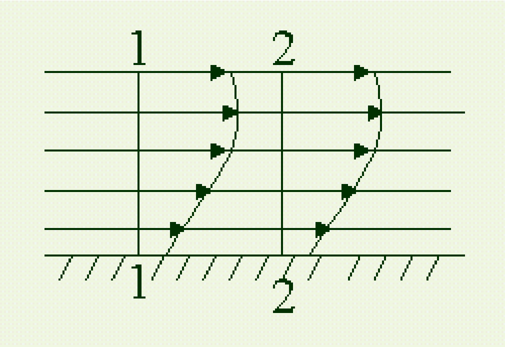
3, Gradually-varied and rapidly-varied flows

4, 1D, 2D, and 3D flow

2, Uniform and non-uniform flows

According to whether the particle motion elements change with the process, it can be divided into:

Uniform flow——Same flow velocity vector at each point along the flow direction, i.e., the migration acceleration $\left(\vec{u}\cdot\nabla\right)\vec{u}=0$



Examples: straight pipe flow of equal cross-section;

water flow in long straight channels

Non-uniform flow——flow velocity vector changes along the flow direction, i.e., the migration acceleration $\left(\vec{u}\cdot\nabla\right)\vec{u}\neq 0$

Example: Fluid flow in a shrink tube

$$u_x = t, u_y = -y, u_z = z$$

Example try to judge: (1) Is it steady flow? (2) Is it uniform flow?

$$\frac{\partial u_x}{\partial t} = 1 \neq 0$$

$$u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} + u_z \frac{\partial u_y}{\partial z} = y \neq 0$$

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2, Euler method for classification of flows

1, Steady and unsteady flows

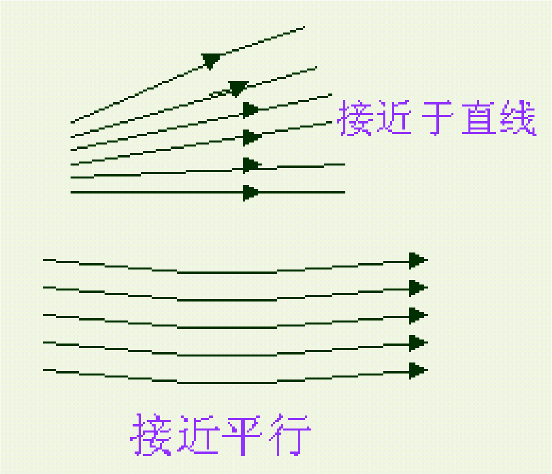
2, Uniform and non-uniform flows

3, Gradually-varied and rapidly-varied flows

4, 1D, 2D, and 3D flow

3, Gradually-varied and rapidly-varied flow

In a non-uniform flow, if the flow changes slowly and the curvature of the streamline is small and close to parallel, it is a gradually-varied flow, otherwise it is a rapidly-varied flow.



Rapidly-varied flow: Flow that changes sharply along its path.

Features: The included angle between the streamlines is large or the radius of curvature is small, or both.

2, Euler method for classification of flows

1, Steady and unsteady flows

2, Uniform and non-uniform flows

3, Gradually-varied and rapidly-varied flows

4, 1D, 2D, and 3D flow

1D flow

1D flow: Flow velocity depends on only one coordinate:

$$\vec{u} = \vec{u}(s,t)$$

Euler acceleration: $\vec{a} = \frac{\mathrm{d}\vec{u}}{\mathrm{d}t} = \frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \frac{\partial \vec{u}}{\partial s}$

Questions:

Euler acceleration for 1D steady flow? $\vec{a} = \vec{u} \cdot \frac{\partial \vec{u}}{\partial s}$

Euler acceleration for 1D uniform flow? $\vec{a} = 0$

2D flow

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2D flow: Flow velocity field can be described by two (not necessarily Cartesian) coordinates.

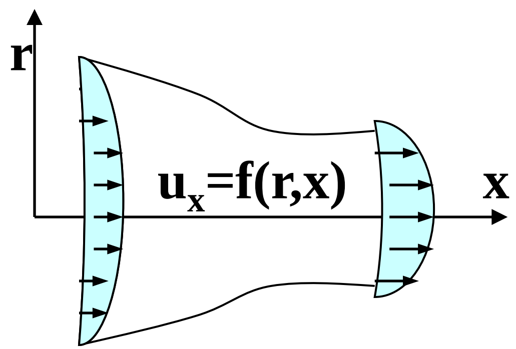


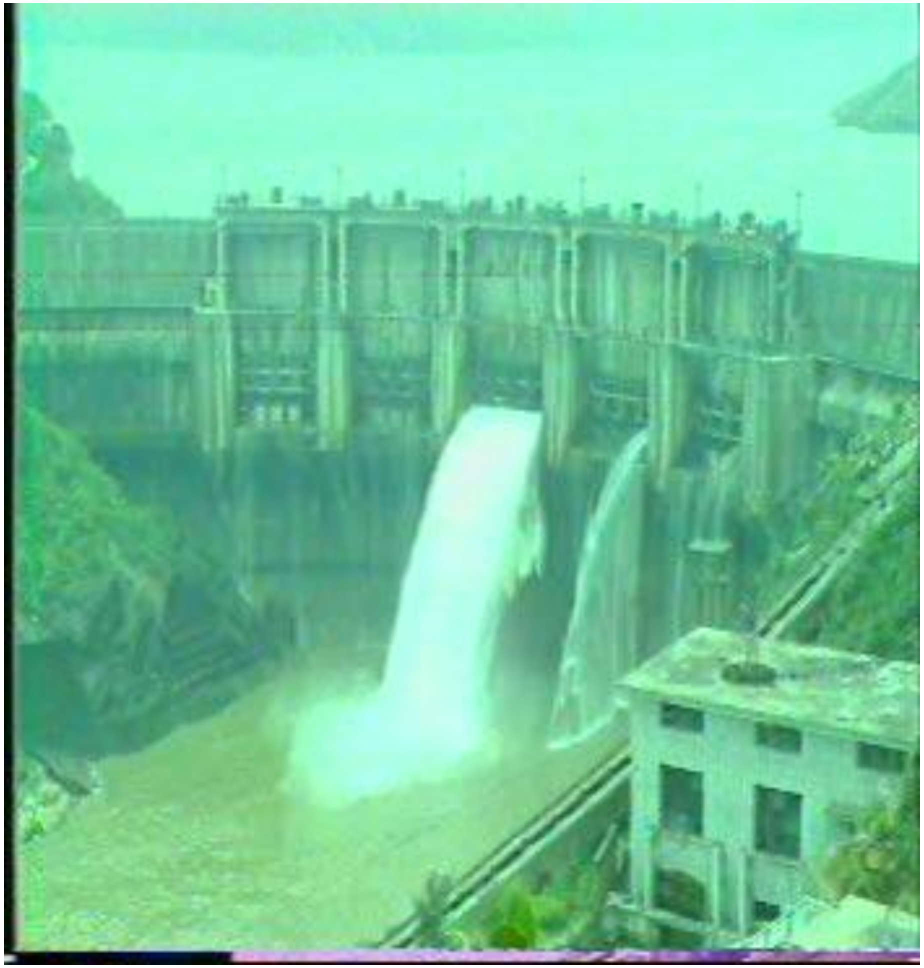
Figure 1

Figure 2

3D flow

3D flow: Flow velocity field varies in all directions

For example, water flows in natural river channels whose cross-sectional shape and size vary along the course; water flows around ships.



Laminar and turbulent flows (*)

(1) Laminar Flow

Laminar flow is characterized by fluid particles following smooth paths in layers, with each layer moving smoothly past the adjacent layers with little or no mixing. Viscosity inhibits the lateral motion of fluid particles across layers.

(2) Turbulent Flow

As the flow velocity increases, the flow layers are more and more unstable, and the fluid particles are mixed with each other and move along a very irregular paths.

Turbulence is very common in engineering practice. Turbulent flow particles can not only undergo lateral fluctuations but also reverse motion relative to the total fluid motion. In turbulent flow, the magnitude and direction of the flow velocity change rather randomly. Turbulence is the reason why the needle of the pressure gauge keeps oscillating.

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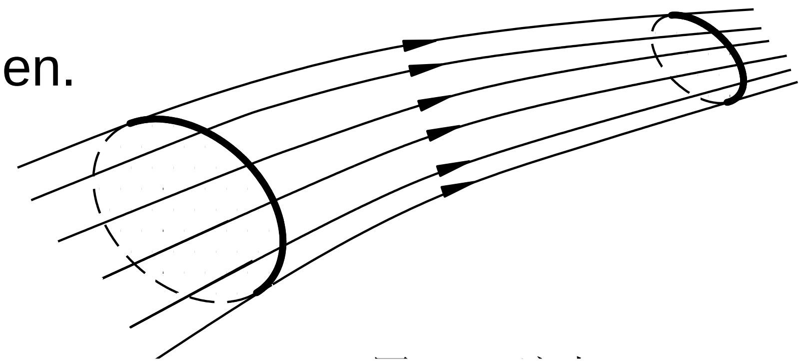
Section 2 Basic concepts of fluid flow

1, Stream tube, stream filament

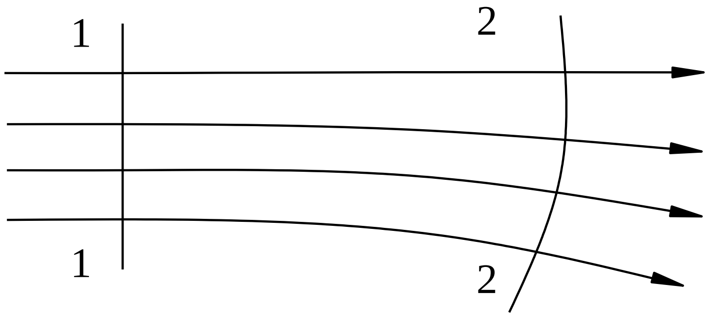
Stream tubes: streamlines through each point of a closed curve and space in between.



Stream Filament: Infinitesimal stream tube

2, Flow cross-section

The section in the flow field that is orthogonal to the streamlines.



3, Discharge

The amount of fluid flowing through a flow section per unit time

$$q_V = \int_A u \mathrm{d}A \text{ (volume flux)}$$

$$q_m = \int_A \rho u \mathrm{d}A \text{ (mass flux)}$$

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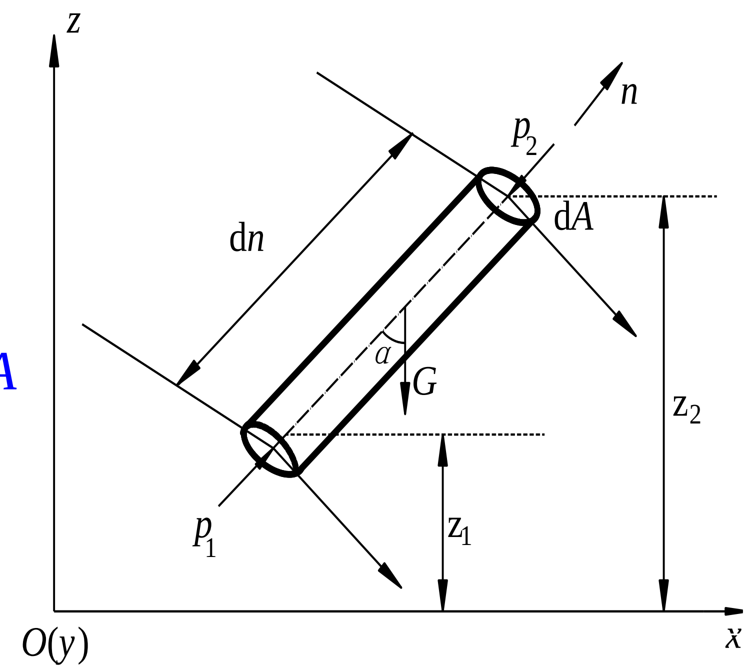
Section 2 Basic concepts of fluid flow

4, Cross sections of gradually-varied flows

1. The cross section of a gradually-varied flow is approximately a plane (***xOz*** in the figure), and the direction of the flow velocity at each point of the section is almost parallel;
2. The fluid dynamic pressure of a gradually-varied flow at the same flow cross-section is approximately distributed according to the law of static pressure:

i.e., constant $z+\dfrac{p}{\rho g}$



Proof:

1, Gravity along ***n***-axis: $-G\cos\alpha=\rho g\mathrm{d}A(z_1-z_2)$

2, Normal force on top and bottom surfaces: $-p_2\mathrm{d}A$、$p_1\mathrm{d}A$

3, Tangential force on cylindrical surface: $F_\tau=0$
(no flow in tangential directions)

4, Total force in ***n***-direction **= 0**

$$\sum F_n=p_1\mathrm{d}A-p_2\mathrm{d}A+\rho g\mathrm{d}A(z_1-z_2)=0$$

$$z_1+\frac{p_1}{\rho g}=z_2+\frac{p_2}{\rho g}=\mathrm{const}$$

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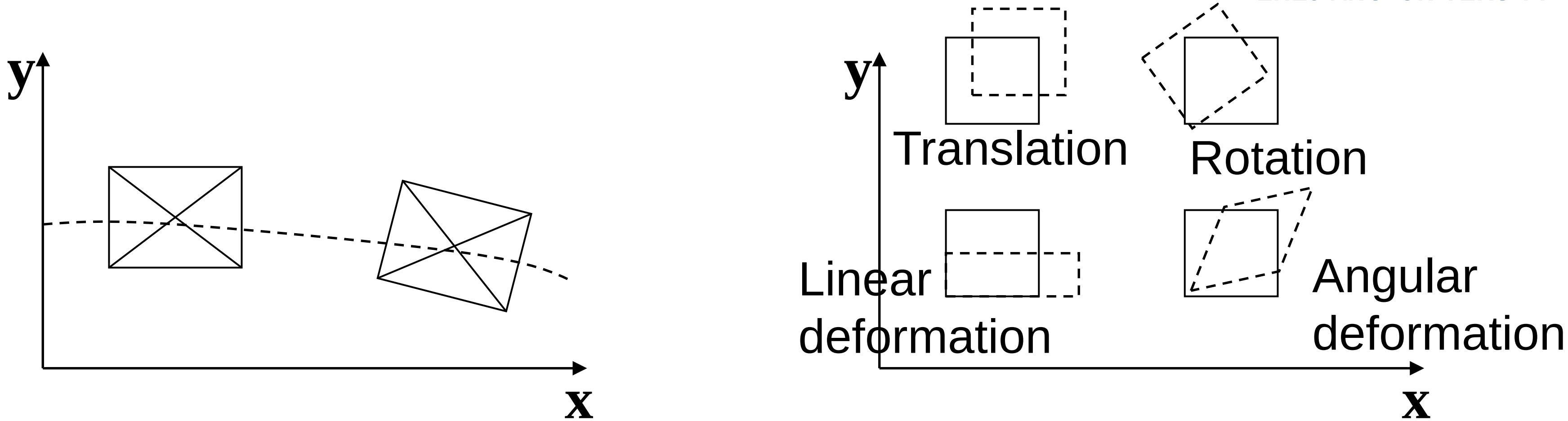
Section 3 Analysis of fluid particle motion

1, Types of fluid particle motion



(a) Rigid body (b) Fluid

The motion of a rigid body is due to translation and rotation about an instantaneous axis.

The movement of a fluid particle generally requires deformation (angular and linear) in addition to translation and rotation.

According to Helmholtz's velocity decomposition theorem, many actual fluid motions are composed of two (or three) basic motion forms of translation, rotation, and deformation (angular and linear).

Section 3 Analysis of fluid particle motion

1, Types of fluid particle motion

2, Angular velocity of fluid particle

3, Rotational and irrotational flows

2, Angular velocity of fluid particle

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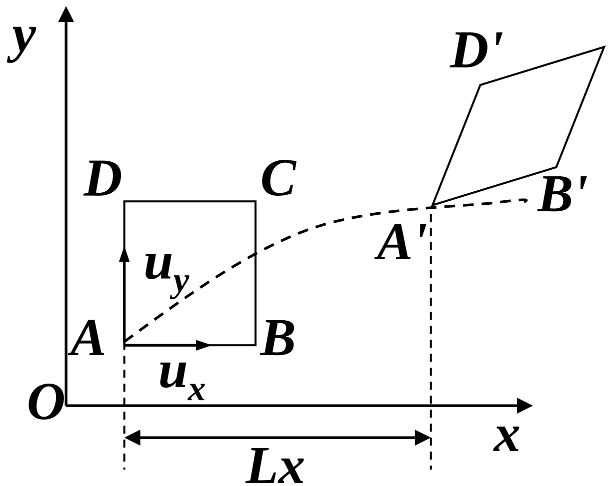
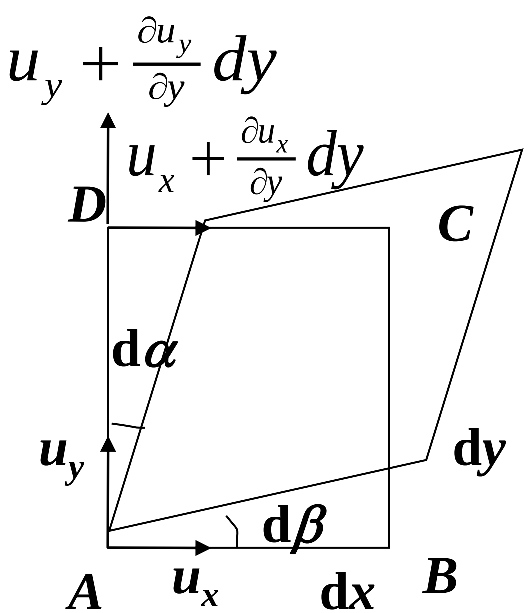
Is defined as the average angular rotation speed

of the two adjacent sides that are perpendicular to each other.



$$\begin{cases} d\alpha \approx \tan d\alpha = \dfrac{\dfrac{\partial u_x}{\partial y}dydt}{dy} = \dfrac{\partial u_x}{\partial y}dt & \text{clockwise (negative)} \\ d\beta = \dfrac{\partial u_y}{\partial x}dt & \text{counterclockwise (positive)} \end{cases}$$

$$\begin{cases} \omega_\alpha = \dfrac{d\alpha}{dt} = -\dfrac{\partial u_x}{\partial y} \\ \omega_\beta = \dfrac{d\beta}{dt} = \dfrac{\partial u_y}{\partial x} \end{cases}$$

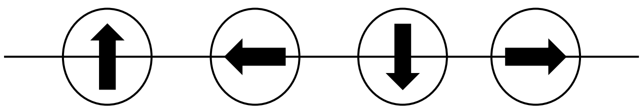


$$\therefore \begin{cases} \omega_z = \dfrac{1}{2}(\omega_\alpha + \omega_\beta) = \dfrac{1}{2}\left(\dfrac{\partial u_y}{\partial x} - \dfrac{\partial u_x}{\partial y}\right) \\ \omega_y = \dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial z} - \dfrac{\partial u_z}{\partial x}\right) \\ \omega_x = \dfrac{1}{2}\left(\dfrac{\partial u_z}{\partial y} - \dfrac{\partial u_y}{\partial z}\right) \end{cases}$$

Section 3 Analysis of fluid particle motion

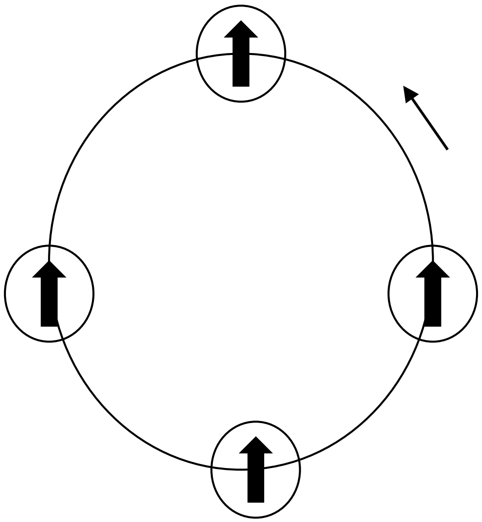
3, Rotational and irrotational flows

Rotational flow (contains vortices)



(a) Rotation

In a vortex, a moving fluid particle not only translates (or deforms), but also rotates with a nonzero angular velocity. A whirlwind is a vortex of air. When the fluid velocity changes greatly, due to the influence of fluid viscous resistance, uneven pressure, and other factors, it is easy for vortices to form.



(b) No rotation

Irrotational Flow, also known as potential flow, does not contain vortices. Fluid particles only translate and deform but do not rotate.

Note: Rotation refers to a fluid particle itself (a) and has nothing to do with its trajectory (b).

Properties of (ir)rotational flows

(1) Potential flow: $\omega_x = \omega_y = \omega_z = 0$, that is,

$$\begin{cases} \dfrac{\partial u_y}{\partial x} - \dfrac{\partial u_x}{\partial y} = 0 \\ \dfrac{\partial u_x}{\partial z} - \dfrac{\partial u_z}{\partial x} = 0 \\ \dfrac{\partial u_z}{\partial y} - \dfrac{\partial u_y}{\partial z} = 0 \end{cases}$$

(2) Rotational flow——at least one of the ω_x, ω_y, ω_z is nonzero.

Example: rotational or irrotational? $u_x = ax, u_y = by, u_z = 0$

$$\omega_x = \frac{1}{2}\left(\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z}\right) = 0$$

$$\omega_y = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} - \frac{\partial u_z}{\partial x}\right) = 0$$

$$\omega_z = \frac{1}{2}\left(\frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y}\right) = 0 + 0 = 0$$

Irrotational.

Summary of basic concepts

1. Lagrangian and Eulerian methods

- Lagrangian: describes the motion of fluid particles through space and time.
- Eulerian: describes locations in space and time through which the fluid flows rather than the motion of the fluid itself
- Euler acceleration: $\vec{a} = \underbrace{\frac{\partial \vec{u}}{\partial t}}_{\text{Local acceleration}} + \underbrace{(\vec{u} \cdot \nabla)\vec{u}}_{\text{Convective acceleration}}$

Summary of basic concepts

2. Streamlines and pathlines

- Streamlines: describe the motion of different particles in the flow field at the same time. The tangent coincides with the flow direction.

do not intersect, are smooth, are denser when the flow velocity is high. $\dfrac{\mathrm{d}x}{u_x}=\dfrac{\mathrm{d}y}{u_y}=\dfrac{\mathrm{d}z}{u_z}$, time $\boldsymbol{t}$ is given

- Pathlines: describe the trajectories of fluid particles as functions of time.

$\dfrac{\mathrm{d}x}{u_x}=\dfrac{\mathrm{d}y}{u_y}=\dfrac{\mathrm{d}z}{u_z}=\mathrm{d}t$, $\boldsymbol{t}$ is an independent variable

Summary of basic concepts

3. Steady and unsteady flows

- Steady flow: The entire flow field does not change with time, and the time-varying acceleration is 0. $\frac{\partial \vec{u}}{\partial t} = 0$

- Unsteady flow: There are spatial locations where the flow field changes with time. $\frac{\partial \vec{u}}{\partial t} \neq 0$

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Summary of basic concepts

4. Uniform and non-uniform flows

- **Uniform flow:** Flow velocity vector is the same along the flow direction, streamlines are parallel straight lines, convective acceleration is 0. $\dfrac{\partial \vec{u}}{\partial s} = 0$

- **Non-uniform flow:** Flow velocity vector is not the same along the flow direction, streamlines are curved lines, convective acceleration is not 0. $\dfrac{\partial \vec{u}}{\partial s} \neq 0$

Summary of basic concepts

5. Gradually-varying and rapidly-varying flows
   - Gradually-varying flow: can be approximated as a uniform flow. The pressure is roughly hydrostatic along the flow cross-section: $z+\dfrac{p}{\rho g}=C$
   - Rapidly-varying flow: convective acceleration cannot be neglected, pressure is not hydrostatic
6. 1D flow and 2D flow
   - 1D flow: flow is function of only one coordinate $\vec{u}=\vec{u}(s,t)$

$$\vec{a}=\frac{\mathrm{d}\vec{u}}{\mathrm{d}t}=\frac{\partial\vec{u}}{\partial t}+\vec{u}\frac{\partial\vec{u}}{\partial s}$$

   - 2D flow: flow is a function of only two coordinates

$$\vec{u}=\vec{u}(r,s,t)$$

Summary of basic concepts

7. Laminar and turbulent flow

- Laminar flow: characterized by fluid particles following smooth layers, with each layer moving past the adjacent layers with little or no mixing.

- Turbulent flow: fluid particles layers are mixed with each other and fluid particles move along a very irregular paths.

Questions

1, What are streamlines, pathlines, and dye lines? How are they different?

Streamlines = tangents coincide with flow velocity direction at given time; pathlines = path of fluid particles throughout time; dye lines = connection of many fluid particles passing through a given point in the past.

2, What are the properties of streamlines and pathlines? What do dye lines do?

Properties of Streamlines:

a. Different streamlines cannot intersect.

b. A streamline is a smooth curve.

c. The density of streamline clusters reflects the size of the flow

Dye lines can be used for flow visualization. In steady flow, the dye lines represent streamlines and pathlines.

3, Are there streamlines in the actual water flow? What is the significance of introducing the concept of streamline?

No. For flow field visualization.

4, "Only when the actual flow velocity of each point of the cross section of the water flow is equal, it is a uniform flow", is this statement correct? Why?

Incorrect. The flow velocity can change across the cross section, only its convective acceleration must vanish.

5, What are the characteristics of steady flow and uniform flow?

Steady flow: $\frac{\partial \vec{A}}{\partial t} = 0$, streamlines = pathlines = dye lines

Uniform flow: $\frac{\partial \vec{u}}{\partial s} = 0$, parallel streamlines, convective acceleration **= 0**

6, What do the Eulerian and Lagrangian Methods describe? What is the method of sitting on a rafting boat to observe the water motion of a river? What is the method of observing water motion from a hydrological station?

Euler: the spatio-temporal space through which the fluid flows, Lagrangian: motion of fluid particles; Lagrangian; Eulerian

7, What are (ir)rotational flows? What are their properties?

Rotational flow: contains vortices. A vortex requires at least one of the ω_x, ω_y, ω_z to be nonzero.

Irrotational flow (= potential flow): no vortices, $\omega_x = \omega_y = \omega_z = 0$ everywhere.